Ali Nawaz

AI Engineer | Python | Machine Learning | Computer Vision | NLP | Gen AI| Agentic AI

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Website

PROFESSIONAL SUMMARY

Dynamic and results-driven AI Engineer with over 5 years of experience transforming complex **business challenges** into innovative AI solutions. Specializing in **Python**, **Machine Learning**, **Deep Learning**, **Data Science**, **Computer Vision**, **NLP**, **PyTorch**, **TensorFlow**, **PostgreSQL**, and **Web Scraping**. Proven expertise in developing scalable and efficient applications by leveraging advanced machine learning algorithms and data science techniques to derive actionable insights. Passionate about driving technological innovation and delivering high-impact results.

PROFESSIONAL EXPERIENCE (4+ Years)

Invozone, Johar Town Lahore, Pakistan (Remote) April 2025 – Present

Senior AI/ML Engineer

Projects:

- **L34H – AI-Powered IT Automation Platform** | *Python, Azure Search/OpenAI, FastAPI, PostgreSQL*
  Developed an intelligent IT automation system integrating which autonomously detect, resolve, and rollback IT issues on Windows systems. Implemented real-time WebSocket communication, LangGraph-based chat interface, Azure Search semantic indexing, and REST APIs.
- **Mynatek – Hospital Cleaning Verification System (Flagship Project)** | *PyTorch, YOLOv11, ST-GCN*
  Led the AI development of an edge-based hospital cleaning verification system combining pose estimation, object segmentation, and action recognition for real-time hygiene monitoring.
  - Built and optimized ST-GCN models for temporal action classification and YOLOv11-based segmentation pipelines for high-touch surface detection.
  - Developed full training and testing pipelines with data augmentation, annotation tools, and visualization modules for AI validation and deployment [Link] .
- **AutoScanX – Real-Time ANPR Prototype** | *PaddleOCR, OpenCV*
  Developed a real-time ANPR system for multi-lane, moving-vehicle environments using IR camera feeds.

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doAZ, Gangnam-gu, Seoul, South Korea (Project Based Contract - Remote) Oct 2024 – March 2025

Senior Computer Vision Engineer

- Led **data labeling** using **Roboflow** for detection and segmentation to refine AI model accuracy.
- Developed a customized **Template Matching** based **Annotator** for creating the Blueprint floor dataset.
- Created the production-ready **Sobang 2.0** and **Sobang 3.0** application for the South Korean Government, enabling automatic symbol counting in civil engineering drawings for building approvals [Link].
- Enhanced **Sobang 3.0** with automatic segmentation, area calculation, and floor number and area detection using **SAM2** and **YOLO** segmentation with customized solutions.
- Endpoints for automatic symbols detection model for civil drawings using **Template Matching** and **YOLOv8**.
- Contributed to the development of **PointChecker**, an AI-powered document analysis tool for automated review, citation, and structured extraction using **LLMs** and **RAG pipelines** [Link].
- Built **PDF parser** with different SOTA parser like Llamaparse, OpenAI, Gemini Flash 2.0, Marker, Docling, markitdown and Cohere Rerank 3.5 for accurate document processing.
- Researched **GAN-based** image background removal and **Real-ESRGAN-based image restoration** to enhance blueprint clarity.

Horizon Tech Services, (NSTP, NUST) Islamabad, Pakistan Dec 2023 – Sep 2024

Python | Lead AI Engineer

- **Developed AI-based Media Content Analysis:** Led the development of a comprehensive AI based Media Content Analysis system, managing electronic, print, and social media data:
  - **Electronic Media**: Directed **low-latency live streaming** for 64+ channels (Urdu/English) with 30 days

backup and built **Clip Factory**, a real-time **video clipping**, tool using **OCR**.

- **Print Media**: Automated **news scraping** from 50+ sources and improved data handling with **Elasticsearch** and **e-newspaper analysis tools**.
- **Social Media**: Developed **Twitter** and **YouTube bots** for content extraction and applied multiple **NLP** algorithms for **video** content summarization, and **sentiment analysis**.

- Built ML pipelines (training & finetuning) for multilingual transcription, translation, OCR, text classification, and sentiment analysis.
- Directed multiple projects using Agile and Scrum methodologies in Jira. Delivered tailored solutions to clients, maintaining strong relationships.
- Supported AI projects with data labeling, model development, and automated annotation using advanced tools, improving team productivity.

Containerized applications for consistent deployment across platforms.

Key Milestones: UDP, TCP Live Streaming, Video on Demand, HLS, ffmpeg, Video and Audio Encoding, Urdu and English OCR, Speech to Text (STT), Text Translation (TTT), Text Summarization, Hugging Face OpenAI, LLM, GitHub

Technology Stack: Python, Django, Machine & Deep Learning, BeautifulSoup, Selenium, Scrapy Langchain, Postgres, Elastic, HPC deployment, GPU Clustering and Parallel Processing

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Horizon Tech Services, (NSTP, NUST) Islamabad, Pakistan — July 2022 – Dec 2023

Python | AI Engineer

- Conducted **R&D** in deep learning and NLP for **sentiment analysis**, **text translation**, and **news classification**.
- Enhanced real-time object detection in Smart Surveillance Systems using **YOLOv8** and **YOLOv10**.
- Collaborated on bird species recognition projects using **Faster R-CNN**, **ResNet50**, and **VGG19**, successfully classifying **55 species** from the **Xeno-Canto** dataset.
- Developed **RRDB** models for image enhancement and **upscaling** of low-resolution images.
- Contributed to aerial imagery projects like **Aerial Imagery** (Uqab) and smart surveillance, focusing on model training and testing.
- Implemented **copy-move forgery** detection in images using **SIFT, SURF, IHC**, and **AKAZE** methods.
- Developed **adversarial attack prevention** mechanisms using generative models and **Defense-GAN** to enhance AI model robustness.

Xylexa, (NSTP, NUST) Islamabad, Pakistan — Dec 2020 – June 2022

AI Engineer

- **Medical Imaging based CAD Development:**
  - Worked on breast cancer detection by analyzing **mammograms** for **lesion identification**.
  - Developed **chest X-ray triage** and multi-disease classification models, including **lung segmentation** for **pneumonia detection**.
  - Utilized **multi-modality** datasets like **MIMIC-CXR, VinDr-Mammo, VinDr-PCXR**, and **VinDr-SpineXR** to enhance model accuracy.
  - Collaborated with radiographers for **annotation** and problem understanding, improving diagnostic precision.
  - Enhanced image quality and updated **Nifti** and **DICOM** tags.
  - **Incremental AI Learning** processes into existing pipelines for training and **finetuning** models.
  - Used **GitHub** for project version control and **Docker** for containerization.
  - Completed medical imaging certifications, including the **CITI Program for Human Research**.

Technology Stack: Python, Computer Vision, NLP, Docker, Sk-learn, Keras, TensorFlow, Pytorch, GitHub, AWS EC2.

University of Engineering and Technology, Taxila, Pakistan — Mar 2019 – July 2020

Research Assistant (TA)

- Start AI research at **SIMPLE Lab** during my Master's degree, focusing on **brain MRI** analysis.
- Developed AI frameworks for medical imaging, with a focus on **Alzheimer's Disease** detection.
- Specialized in **2D** and **3D** imaging techniques, applying AI to improve diagnostic accuracy.

- Explored innovative imaging technologies and their AI integration for enhanced medical analysis.

INTERNSHIPS(1.5 Years)

DYNIMAX Intermedia, Islamabad, Pakistan Jan 2017 – June 2017
Internship : KP IT Board Internship Program

- Engaged in the **KP IT Board Internship** Program.
- Mastered in graphic design tools: **Photoshop**, **Illustrator**, **Corel Draw**, Sony Vegas, and Canva.
- Enhanced skills in graphic design, video editing, photography, and 2D/3D animation.

Cambridge Model College, Karak KPK, Pakistan Mar 2017 – Mar 2018
KP IT Board Internship Program
Internship : National Internship Program (NIP)

- Participated in the **National Internship Program** at CMC Karak.
- Collaborated with **KPK IT Board Peshawar**, contributing to software industry projects.
- **Senior ICT Instructor** teaching computer science and overseeing labs at Cambridge Model College, Karak

EDUCATION

University of Engineering and Technology – Taxila, Pakistan 2017 – 2020
MS - Software Engineering.
CGPA - 3.21/4.00
Thesis Title: Deep Convolutional Neural Network-based Classification of Alzheimer's disease using MRI Data

Major Subjects: Advance Machine Learning, Software System Quality, Research Methodology, Advanced Software Technologies, Algorithm Design and Analysis, Software System Architecture.

Relevant Skills: Programming, Software Development, Data Structures & Algorithms, Research Methodology, Machine Learning, AI.

University of Science and Technology – Bannu, Pakistan 2012 – 2016
B.Sc. - Software Engineering.
CGPA - 3.53/4.00
FYP Title: Online Job Portal

Major Subjects: C Programming, C++, Linear Algebra, Calculus, Statistics, and Probability Theory, Software System Quality, Software Engineering, Discrete Mathematics, Compiler Construction, Artificial Intelligence, Database Systems

Relevant Skills: Procedural Programming, Data Structures & Algorithms, C#, Desktop Application, Database

Cambridge Model College HSSC – Karak, Pakistan 2009 – 2011
F.Sc. – Pre-Engineering.
Marks - 724/1100

CORE SKILLS

- **Languages:** C, C++, Python
- **Frameworks:** Flask, Fast API, Pytorch, TensorFlow
- **Databases:** Postgres, SQLite
- **Real-time Streaming:** RTMP, UDP, TCP, ffmpeg, h264
- **API's:** Fast API
- **Research & Technical Writing:** MS 365, Overleaf, Latex
- **Project & Task Management:** Jira, Agile, Scrum, Trello,
- **AI:** Machine & Deep Learning, CV, NLP, LLM, OpenAI
- **Front End:** HTML, HTML5, CSS, Bootstrap
- **Cloud & Deployment:** AWS EC2, AWS (Intermediate)
- **Unit testing:** Python Unit Test, Pytest
- **API's Testing:** Thunder client, Postman
- **Operating System:** Windows, Linux, and Window Servers
- **Graphic Designing:** AP, AI, Lite Room, CorelDraw, Canva

- **Source Control & Continuous Integration:** GitHub, Gitlab, TFS via Azure DevOps Server, GitHub Actions
- **Others:** Json, Xml, Wrote Documentation of Complex for Operational Teams, Leadership Skills, Self-starter, Collaborative, Organized

AWARDS AND ACHIEVEMENTS

- **Prime Minister Laptop Scheme Award**: Recognized for ranking in the top ten in Software Engineering at the University of Science and Technology Bannu.
- **Diya Foundation Scholarships**: Awarded for academic merit and financial need during bachelor's studies.
- **Best Student Award**: Honored for excellence in academics and extracurricular activities.
- **Fuji Foundation Scholarship**: Granted to beneficiaries for outstanding performance.

LEADERSHIP AND VOLUNTEERING

- **AI Seminars:** Spearheaded sessions at universities on AI, Machine Learning, and Data Science, engaging students and professionals.
- **Online Tutoring:** Volunteered to mentor underprivileged students in tech education and career development.
- **Community Leadership:** Directed a community of 550+ students, organizing workshops and coding sessions for skill enhancement.
- Collaborated on transformative projects "**Opti Key**" for blind persons to utilize technology, and "**Bacteria Cell Counter**" heralding advancements in assistive technology for biotechnology field.

PUBLICATIONS

Journals:

Nawaz, Ali et al., "Weakly-Supervised Region Proposals for Chest-Xray Disease Localization and Classification," IEEE Transactions on Medical Imaging, 2023. (Submitted)

Manzoor, Sohail et al., "Melanoma Detection Using a Deep Learning Approach," International Journal of Innovations in Science & Technology, 2022.

Conference:

Nawaz, Ali et al., "Deep Convolutional Neural Network-Based Classification of Alzheimer's Disease Using MRI Data," IEEE INMIC, 2020.

Khan, et al., "Optimized Prediction Model for Type 2 Diabetes Mellitus Using Gradient Boosting Algorithm," MAJICC, 2021.

Iqbal, Javid et al., "Ordinary Learning Method for Heart Disease Detection Using Clinical Data," iCoMET, 2020.

Ahmad, Fawad et al., "Mortality Prediction in ICU Patients Using Machine Learning Models," IBCAST, 2021.

Nawaz, Ali, Syed Muhammad Anwar, Javed Ali Khan, Weakly-Supervised Region Proposals for Chest-Xray Disease Localisation and Classification (IEEE Transactions on Medical Imaging (T-MI).

REFERENCES

Academic:

Assoc. Prof. Engr. **Dr. Syed M. Anwar**, Medical Image & Signal Processing, University of Engineering and Technology, Taxila (UET), Pakistan. Email: s.anwar@uettaxila.edu.pk Contact: +92-300-5044036, Faculty Profile Page: [Link]

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